BUDGETED CORRECTION IN GYNECOLOGIC VIDEO SEGMENTATION AN EMPIRICAL AUDIT OF SELECTION AND LEARNED VERIFICATION

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ABSTRACT

We audit budgeted full-mask correction on 170 AutoLaparo training clips from nine source videos using a frozen framewise model and SAM 2.1 Tiny. Initialization is charged and query/scoring frames are disjoint. Raw entropy raises source-macro Dice from 0.8035 to 0.9150 with five masks, but 43 clips lose Dice between two and five masks. Source-grouped acceptance and a separately registered utility selector fail their internal promotion criteria. Ten feature/fallback ablations also remain below raw in the original scoring pool. A full-cohort close-control extension collects all 170 outputs but fails historical raw-bank parity on 20 clips; its negative adaptation results are retained in a nonconforming exploratory appendix. On one diagnostic clip, disabling BF16 autocast reduces cross-class drift without ensuring Dice gains. These findings support transparent correction accounting and reproducibility checks, not a superior allocator or clinical benefit.

***Index Terms*—** Surgical video, segmentation, full-mask correction, source grouping

1. INTRODUCTION

Video segmentation can assist annotation of laparoscopic procedures, but propagating an additional correct mask can change predictions unfavorably elsewhere. Evaluating allocation therefore requires matched correction units, initialization, query access, scoring frames and video grouping.

Active annotation and reliable propagation have close prior work. EVA-VOS learns quality representations and annotation-type selection [1]. SASVi automatically re-prompts tracking from a framewise overseer [2]. These systems differ in supervision and intervention unit: a predicted prompt is not a newly supplied human mask, and tracking scores cannot be compared directly with charged-correction AUC. We do not claim uncertainty, diversity, re-prompting or learned quality prediction as novel in themselves.

We test whether purchased masks verify propagation beyond observable controls, followed by one bounded utility selector, on official sparse AutoLaparo references [3]. Both remain negative development findings. A gold-aware finite-space diagnostic and a failed historical-parity extension delimit their interpretation. The contribution is an empirical audit, not a superior allocator.

2. DATA AND EVALUATION

All 300 Task 2 clips were assigned to 16 source videos using six-frame RGB correspondence, alternative-source margins, monotonic alignment and agreement with official counts, pools and durations. These are video groups, not proven patient identities. We analyze the 170 official training clips from nine sources. The validation and final-test partitions contain 57 and 73 clips; neither learned candidate was promoted to a new validation or final-test run.

Task 3 supplies references at six zero-based frame indices: 0, 24, 49, 74, 99 and 124. We use 125 raw frames at 25 fps, query frames $[0, 24, 49]$ and disjoint scoring frames $[74, 99, 124]$, identical for every policy. Targets are uterus and the union of release-annotated primary instrument parts. Encoding and alignment checks do not constitute clinical adjudication or exhaustive labeling of every tool. Missing tail annotations are never treated as negative references.

The frozen DINOv2-based framewise model [4] has an audited training manifest of 5,955 rows: 4,992 instrument, 725 official anatomy and 238 verified anatomy images from related gynecologic work [5]. No explicit AutoLaparo training row or exact Task 3 file match was found for this artifact. Nevertheless, earlier related benchmarking and repeated development exposure prevent describing these sources as pristine unseen cases. Foundation-model pretraining overlap remains unknown.

One correction purchases a full binary mask of one target at one query frame, including initialization. Budget zero uses only framewise probabilities. Fixed SAM 2.1 Tiny [6], BF16 inference and disabled postprocessing propagate paid masks offline in both directions, independently per target. Uncorrected targets retain framewise predictions. Within-clip contrasts share models, resolution, candidate pools, scoring pools and hardware.

We purchase 0, 1, 2 or 5 masks, giving actual densities of 0, 0.8, 1.6 or 4 masks per 100 raw frames, respectively. Dice and IoU average the two targets and three scoring frames, clips within source, then sources equally. Jointly empty reference and prediction score one; reference-present sensitivity

is separate. Normalized trapezoidal Dice–density AUC is

$$A = 0.1D_0 + 0.2D_1 + 0.4D_2 + 0.3D_5. \quad (1)$$

Random seeds average within clip before source aggregation. Descriptive paired intervals use 2,000 source bootstrap resamples. Frames and clips are not independent patients, and historical exposure plus multiple exploratory comparisons preclude confirmatory superiority claims.

3. MATCHED SELECTORS AND VERIFICATION

Raw entropy selects the unused frame–target pair with highest mean normalized binary entropy in its current raw state. Uniform prioritizes target balance and temporal spread; random uses seeds 100–109. These are matched selectors, not reproductions of complete prior systems. For development only, three candidates per head give seven nonempty prompt subsets. Fourteen independent propagations plus baseline form a factorized bank. Observable access is restricted to states conditioned on masks already purchased. No selector receives scoring references, metrics or future paid states. This computational reuse is not a latency measurement.

Acceptance compares each scoring output with the last retained output using entropy, disagreement, area, temporal consistency and paid-mask distance. Anchor evidence reconstructs purchased query masks using leave-one-prompt-out states: a prediction hard-overwritten by the same mask cannot verify that mask. With fewer than two masks for a head, support is unavailable and the fixed fallback accepts updates. Labels are quality differences on training references, not inference features.

Anchor and observable-only models share outcome supervision, source-balanced weights and tuning budget. Leave-one-source-out evaluation nests three-fold grouped selection among fixed Ridge/logistic families and two registered thresholds per family. The raw query trajectory persists when an output is rejected. Additional reconstruction propagations cost compute but do not purchase more masks.

After this acceptance result, a separate protocol fixed one source-balanced Ridge utility model ($\alpha = 10$), without hyperparameter search. It predicts marginal gain using current paid-state query entropy and target/time/history features. A static target/time ablation has identical supervision. There are 4,080 action pairs; each held-out source uses the other eight for fitting. This sequentially introduced branch is exploratory, not training-free. Promotion requires at least $+0.005$ mean AUC and six of nine positive sources: an internal development rule, not a significance threshold.

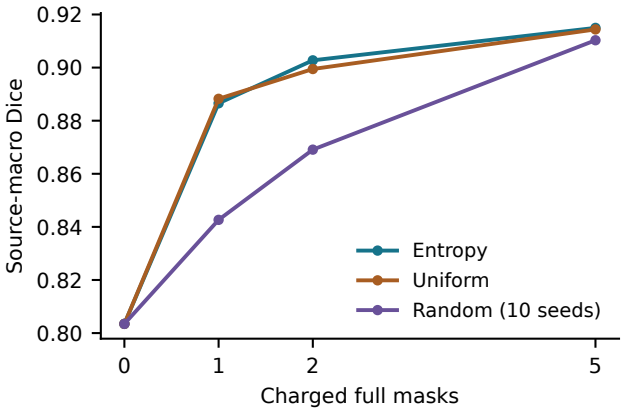


Fig. 1. Matched fixed selectors. All full-mask prompts are charged and no scoring snapshot is a query frame.

4. RESULTS

4.1. Budget gains and non-monotonic corrections

Raw Dice rises from 0.803468 without masks to 0.886549, 0.902732 and 0.914955 with one, two and five masks (Fig. 1). The first mask accounts for 74.5% of the mean zero-to-five gain. This is efficiency in mask counts, not measured clinician time. Entropy exceeds random by 0.023632 AUC, while its difference from uniform remains uncertain (Table 1).

Between two and five masks, 43/170 raw trajectories lose Dice, including 13 decreases greater than 0.02. Anchor acceptance reduces these counts to 40 and 10 while slightly lowering mean AUC. Avoiding some bad updates can also reject useful updates; fewer harmful transitions alone do not demonstrate better allocation or patient-harm reduction.

4.2. Neither learned branch passes promotion

Paid-anchor acceptance changes AUC by -0.000431, with three positive sources. Anchor-minus-observable control is -0.0000458, with interval crossing zero: purchased-mask features have not demonstrated added predictive value. Utility gains 0.001480 on average, but its interval crosses zero and only five sources improve (Fig. 2). Full-minus-static utility is 0.002333, interval $[-0.001137, 0.006517]$. Neither branch is promoted; thresholds, source subsets and model families are not changed to rescue outcomes.

At five masks, raw uterus Dice is 0.847946 on reference-present snapshots versus 0.939573 for the release-annotated tool union. Source counts and empty-reference conventions differ by stratum. These scores do not establish exhaustive tool detection or clinical absence.

Table 1. Completed development results. Differences use raw entropy as reference; intervals are descriptive paired source bootstraps. Both learned branches fail the internal promotion rule.

Method	Dice AUC	Difference	95% source interval	Positive sources
Raw entropy	0.893236	–	–	–
Uniform coverage	0.892092	−0.001144	[−0.006896, 0.004109]	3/9
Random, ten seeds	0.869604	−0.023632	[−0.026483, −0.020431]	0/9
Observable acceptance, OOF	0.892851	−0.000385	[−0.001556, 0.000576]	4/9
Paid-anchor acceptance, OOF	0.892805	−0.000431	[−0.001679, 0.000549]	3/9
Static role/time utility, OOF	0.892383	−0.000853	[−0.006717, 0.004302]	4/9
Entropy utility, OOF	0.894716	0.001480	[−0.001520, 0.004449]	5/9

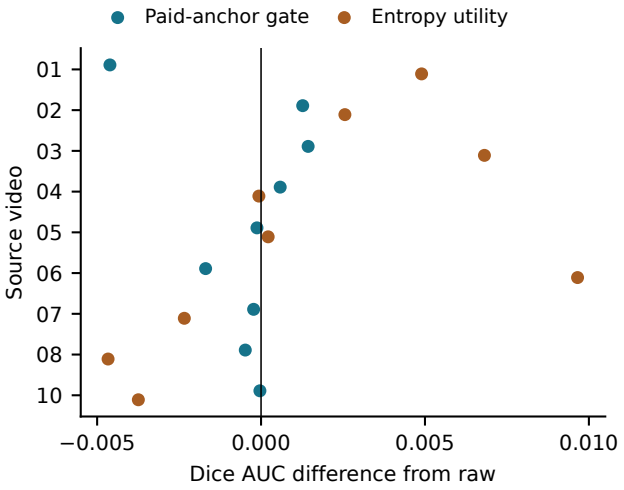


Fig. 2. Learned-branch source effects versus raw. Points are video-source means, not independent per-source confidence intervals.

4.3. Retrospective finite-space headroom

A separately registered evaluator-only diagnostic enumerates 720 ordered sequences of five distinct masks from six actions using existing bank states, without new propagation or fitting. Best-sequence selection per clip using scoring gold yields source-macro AUC 0.920028 versus raw 0.893236, gap 0.026792 (descriptive interval $[0.016340, 0.038904]$). This oracle accesses forbidden future outcomes, and its advantage is partly nonnegative by construction. It bounds only this finite action space, not other prompts, backbones, output blending or tracking systems.

Exact random AUC is 0.872149, below raw by 0.021087. Thus the direction of the raw–random observation is not confined to ten sampled seeds. This posthoc diagnostic does not replace the registered comparator. Remaining gold-aware headroom suggests that a lack of favorable trajectories alone cannot explain the learned branches' failure; it does not establish observable or learnable headroom.

4.4. Reproducibility and compute

Eight A10G banks failed the fixed reference checks. Preserved failures and a bounded exact-A10 technical repair yield 170 accepted banks (164 A10, six A10G), hardware matched within clip. Numerical reconstruction is accurate to 10^{-12}. Bank construction totals 7.564 worker-hours for 14 propagations per clip, excluding some startup/failure overhead; it is neither billing nor policy latency. Existing baseline/JPEG preparation is excluded from the separate sequential benchmark below; mask copies are not clinician time.

4.5. Descriptive replication outside gynecology

A frozen protocol evaluates 101 CholecSeg8k windows from 17 previously explored video sources [7], not blinded validation. Each 80-frame window has queries 0/15/31 and disjoint scores 47/63/79. Budgets 0/1/2/3 correspond to densities 0/1.25/2.5/3.75 per 100 frames. Scoring uses the released grasper/L-hook union, excludes void and differs from the model's broader tool head. Query void is mechanically zero-filled. No failed learned candidate is retuned or evaluated.

Entropy AUC is 0.916173 versus uniform 0.914145 and random 0.904317. Source-equal paired intervals from 10,000 draws give entropy-minus-random 0.011855 $[0.001561, 0.024806]$, with 10/17 positive sources; entropy-minus-uniform is 0.002028 $[-0.005708, 0.012754]$, with 4/17 positive sources. All selectors converge after three actions, yet 30/101 entropy windows lose Dice from two to three masks, including two declines above 0.02. This 3.16-second one-target grid cannot establish long-term tracking or be pooled with AutoLaparo AUC.

4.6. Completion diagnostics and failed historical parity

A post-negative diagnostic uses source-balanced Ridge ($\alpha = 10$, threshold zero), leave-one-source-out fitting and raw queries. Five feature groups cross retained-output versus baseline fallbacks. All ten original-pool means are below raw on 170 clips. Retained output has a larger mean

for every group; the full-feature difference is 0.009471 $[0.005524, 0.014193]$. Fallback changes later features, precluding a causal feature claim or promotion.

We collected raw, EVA adaptation using official QNet with mask prompts and SAM2, and a SASVi-inspired semantic predicted-prompt control on all 170 clips, reusing nine representatives. Modes share hardware within each clip; neither adaptation reproduces the original system. Every gold mask is charged; the automatic control uses zero gold and a separate endpoint. However, the registered historical raw-bank parity gate fails on 20/170 clips, all on A10G, including three query-history differences. Across 127 A10 and 43 A10G clips, all input/output hash, scope, budget and finite-value checks pass. The largest raw Dice/IoU discrepancy is 0.2290; this is not dismissed as negligible roundoff.

The original gate stays failed. A separately documented, nonconforming exploratory appendix retains all 170 without rerunning or dropping failures. EVA AUC is 0.889018 versus contemporaneous raw 0.892098, difference -0.003079 $[-0.007819, 0.001327]$, with 3/9 positive sources. Automatic Dice is 0.791402 versus zero-mask baseline 0.803468, difference -0.012065 $[-0.026281, 0.001406]$. These remain nonconforming exploratory results. An initial diagnostic completed 19/48 A10 cells with 96 exact saved-step pairs. A separately registered allocator-observed panel completed 24 cells (20 A10, four A10G) on six training clips. All 192 same-class scoring-step pairs were exact, yet fixed-query cross-class differences reached 0.216309; routing alone is insufficient. A further counterbalanced precision diagnostic attempted four paired processes on clips 132/148, with TF32 disabled: three completed (six/eight arms), one stopped at a no-overwrite guard, with partial states preserved. Each pair shared a physical GPU and queries. On clip 132, cross-class maximum probability drift was 1.0 under BF16 and 0.000732 without BF16 autocast; the latter remained nonidentical and had lower early-budget Dice on A10. SAM2 still stores mask memory in BF16 internally. These purposive diagnostics neither isolate hardware causality nor repair the original gate. Validation remains protocol-only and final test closed.

5. DISCUSSION AND LIMITATIONS

Mask accounting reveals diminishing returns and per-clip degradation; uniform remains a strong control. Neither learned branch passes promotion. These findings establish neither a successful allocator nor novelty over close prior work.

Nine previously observed development sources are not 170 independent procedures. Nested OOF does not erase design exposure. Sparse five-second windows do not measure dense or long-term tracking; offline lookahead is not causal surgical guidance. References lack new clinical adjudication, and the previously explored external release is not pristine validation. Final test remains closed. Code and aggregates: `https://github.com/nhynhi23411/gynsurg_isbi`; licensed media and weights are excluded.

6. COMPLIANCE WITH ETHICAL STANDARDS

Released data are used for academic noncommercial research without new human annotations. Institutional ethics confirmation and publication permissions remain pending; provider statements do not establish this study's approval or waiver. No new clinician review, safety outcome or annotation-time measurement is claimed.

7. ACKNOWLEDGMENTS

Author identities, affiliations, funding and conflicts are pending. OpenAI Codex assisted analysis software, figures, abstract and manuscript drafting. Recorded numerical checks do not replace author review; this draft is unapproved and unsubmitted.

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